Started on	Monday, 20 December 2021, 2:13 PM
State	Finished
Completed on	Monday, 20 December 2021, 2:30 PM
Time taken	16 mins 9 secs
Grade	**11.67** out of 21.00 (**56%**)

Question 1

Partially correct

Mark 4.67 out of 7.00

Given the following table, the execution of the processes is using **Round Robin (RR)** scheduling algorithm with the time quantum set to 3. **[7 marks]**

Process	Arrival Time	CPU Cycle
P1	0	4
P2	1	6
P3	2	3
P4	3	5
P5	4	7

a) Fill in the cells by choosing which processes (**P1, P2, P3, P4, P5**) that is appropriate . [5 marks]

P1 ✔	P2 ✔	P3 ✔	P4 ✔	P5 ✔	P1 ✔	P2 ✔	P4 ✔	P4 ✖	P2 ✖	
0	3	6	9	12	15	16	21	24	25	(ms)

b) Calculate the Average Waiting Time (AWT) and the Average Turnaround Time (ATT). [2 marks]
[Note: Answer in one decimal point only (E.g. : 5.6) and for whole number there is no need for a decimal point (E.g. : 2)]

Average Waiting Time =

3

✖ ms

Average Turnaround Time =

3

✖ ms

Question 2

Partially correct

Mark 4.00 out of 8.00

Carefully study the following **Peterson's Algorithm** as shown in the following table. There are two processes, **P0** and **P1,** who wish to satisfy the Critical Section requirements. Initially both flags, flag[0] and flag[1] are set to FALSE. Fill in the cells below with the appropriate values and actions. [8 marks]

Process P0	Process P1

while(TRUE)
{
flag[0] = TRUE;
turn = 1;
while (flag[1] && turn == 1)
; /* do nothing */
critical section
flag[0] = FALSE;
print("Hai");
}
	while (TRUE)
	{
	flag[1] = TRUE;
	turn = 0;
	while (flag[0] && turn == 0)
	; /* do nothing */
	critical section
	flag[1] = FALSE;
	print("Good");
	}

[Note :Answer **MUST** use TRUE or FALSE in capital letter].
Put cursor in the first cell and use TAB to go to next cells.

Time	flag[0]	flag[1]	Turn	Events
t0	FALSE	FALSE	0	Both processes P_0 and P_1 alive
t1	TRUE ✗	TRUE	0 ✓	P_1 wishes to enter CS
t2	TRUE ✗	TRUE	0 ✗	P_1 in the CS
t3	TRUE ✓	TRUE	0 ✗	P_0 wishes to enter CS
t4	TRUE ✓	TRUE	0 ✗	P_0 busy waiting
t5	TRUE ✓	FALSE	0 ✗	P_1 leave CS
t6	TRUE ✓	FALSE	0 ✗	P_0 enter CS
t7	TRUE ✓	TRUE	0 ✓	P_1 wishes to enter CS
t8	TRUE ✗	TRUE	0 ✓	P_0 leave CS

Question 3

Partially correct

Mark 3.00 out of 6.00

The following table shows three processes **P1, P2** and **P3** run concurrently in a Unix system. Assume that all the flags, A, B, and C are initially initialized to 0. There are TWO possible output (refer **Table A** and **Table B**) you need to complete both table. [6 marks]

P_0	P_1	P_2
wait(B) print ("B") signal(C)	signal(C) print ("A") signal(B)	wait(C) print ("C")

Output Table A [3 marks]

[Note: For column A, B and C, update the cells with value either **0** or **1**].

Process	A	B	C	Output
Initial	0	0	0	None
P1	1 ✗	1 ✓	1 ✓	A ✓
P0	1 ✗	1 ✗	1 ✗	B ✓
P2	1 ✗	1 ✗	1 ✓	C ✓

Output Table B [3marks]

[Note: For column A, B and C, update the cells with value either 0 or 1].

Process A	A	B	C	Output
Initial	0	0	0	None
P1	1 ✗	1 ✓	1 ✓	A ✓
P0	1 ✗	1 ✓	1 ✗	B ✗
P2	1 ✗	1 ✗	1 ✓	C ✗

Previous activity

◀ Lab 2 Submission

Jump to...

Next activity

GROUP PROJECT ▶

Stay in touch

UTM Academic Leadership - UTMLead

http://utmlead.utm.my/

Mobile : +6075537914

elearning@utm.my

Data retention summary

Get the mobile app